

Sets in Python

A **Set** is a collection of unique elements.

Sets are:

- Unordered
 - Unindexed
 - Mutable (can change)
 - No duplicate values allowed
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Creating a Set

Code

```
numbers = {10, 20, 30}
```

```
print(numbers)
```

Output

```
{10, 20, 30}
```

Duplicate Values Not Allowed

Code

```
numbers = {10, 20, 10, 30}
```

```
print(numbers)
```

Output

```
{10, 20, 30}
```

Duplicate 10 is automatically removed.

Empty Set

Wrong Way

```
a = {}
```

```
print(type(a))
```

Output

```
<class 'dict'>
```

Correct Way

Code

```
a = set()
```

```
print(type(a))
```

Output

```
<class 'set'>
```

Accessing Set Elements

Sets are unordered, so indexing is not possible.

Wrong Code

```
numbers = {10, 20, 30}
```

```
print(numbers[0])
```

Output

```
TypeError
```

Loop Through Set

Code

```
numbers = {10, 20, 30}
```

```
for n in numbers:
```

```
    print(n)
```

Output

```
10
```

```
20
```

```
30
```

(Note: Order may change)

Adding Elements

1. add()

Adds one element.

Code

```
numbers = {10, 20}
```

```
numbers.add(30)
```

```
print(numbers)
```

Output

```
{10, 20, 30}
```

2. update()

Adds multiple elements.

Code

```
numbers = {10, 20}
```

```
numbers.update([30, 40])
```

```
print(numbers)
```

Output

```
{40, 10, 20, 30}
```

Removing Elements

1. remove()

Removes specific element.

Code

```
numbers = {10, 20, 30}
```

```
numbers.remove(20)
```

```
print(numbers)
```

Output

```
{10, 30}
```

Important

If value not found → error occurs.

2. discard()

Removes element safely.

Code

```
numbers = {10, 20, 30}
```

```
numbers.discard(50)
```

```
print(numbers)
```

Output

```
{10, 20, 30}
```

No error occurs.

3. pop()

Removes random element.

Code

```
numbers = {10, 20, 30}
```

```
numbers.pop()
```

```
print(numbers)
```

Output

(Random output)

{20, 30}

4. clear()

Removes all elements.

Code

```
numbers = {10, 20, 30}
```

```
numbers.clear()
```

```
print(numbers)
```

Output

```
set()
```

Set Operations

1. Union |

Combines sets.

Code

```
a = {1, 2, 3}
```

```
b = {3, 4, 5}
```

```
print(a | b)
```

Output

```
{1, 2, 3, 4, 5}
```

2. Intersection &

Common elements.

Code

```
a = {1, 2, 3}
```

```
b = {2, 3, 4}
```

```
print(a & b)
```

Output

```
{2, 3}
```

3. Difference -

Elements in first set only.

Code

```
a = {1, 2, 3}
```

```
b = {2, 3, 4}
```

```
print(a - b)
```

Output

```
{1}
```

4. Symmetric Difference ^

Elements not common in both.

Code

```
a = {1, 2, 3}
```

```
b = {2, 3, 4}
```

```
print(a ^ b)
```

Output

```
{1, 4}
```

Membership Operators**Code**

```
numbers = {10, 20, 30}
```

```
print(20 in numbers)
```

```
print(50 in numbers)
```

Output

True

False

Length of Set

Code

```
numbers = {10, 20, 30}
```

```
print(len(numbers))
```

Output

3

Frozen Set

Immutable set.

Code

```
a = frozenset([1, 2, 3])
```

```
print(a)
```

Output

```
frozenset({1, 2, 3})
```

Important Set Functions

Function	Purpose
add()	Add one element
update()	Add multiple elements
remove()	Remove element
discard()	Remove safely
pop()	Remove random element
clear()	Remove all
union()	Combine sets

Function	Purpose
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intersection()	Common elements
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Difference Between List, Tuple, and Set

Feature	List	Tuple	Set
Symbol	[]	()	{}
Ordered	Yes	Yes	No
Mutable	Yes	No	Yes
Duplicates Allowed	Allowed	Allowed	Not Allowed
Indexed	Yes	Yes	No

Most Important for Exams

Usually asked:

1. Definition of set
2. Difference between list, tuple, set
3. add() vs update()
4. remove() vs discard()
5. Union and intersection
6. Duplicate removal
7. Frozen set

Quick Revision

Property	Set
Ordered	No
Indexed	No
Mutable	Yes
Duplicates allowed	No
Uses {}	Yes